# UNL SoC Falkinburg Career Explore XR

# Senior Design Showcase

## April 24, 2026

# Company

- 4 Majors: Computer Science, Computer Engineering, Software Engineering, and Data Science
- Nearly 1000 Undergraduate Students
- Variety of hands-on learning experiences inside and outside the classroom
- Opportunities to learn from world-renowned faculty, contribute to real-world projects, and connect with classmates and industry professionals

SCHOOL OF COMPUTING

# Stakeholders

Sponsors

Principal Investigator
Jeff Falkinburg

Subject Matter Expert
Britney Falkinburg

# Stakeholders

Faculty, Staff and Coach

Project Manager
Roland Madadjim

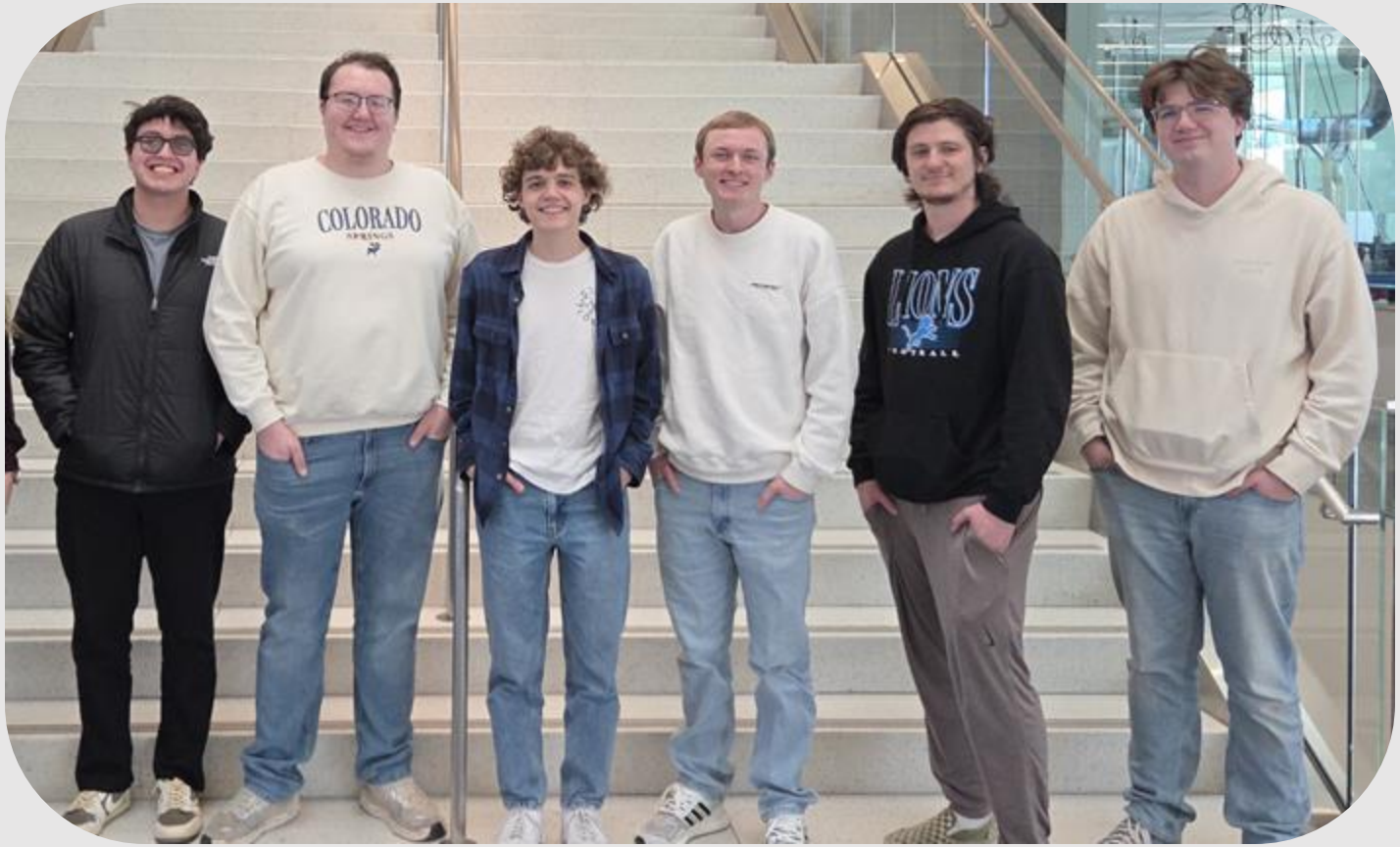
Tribe Lead
Jeff Falkinburg

Industry Coach
Tanner Hughes

# Stakeholders

Student Team

- Will Bernal (Developer)
- Sean Casey (Squad Lead, Product Manager)
- Colman Scharff (Developer)
- Tyson Veik (Developer)
- Noah Russell (Developer)
- Burke Groenjes (Development Manager)



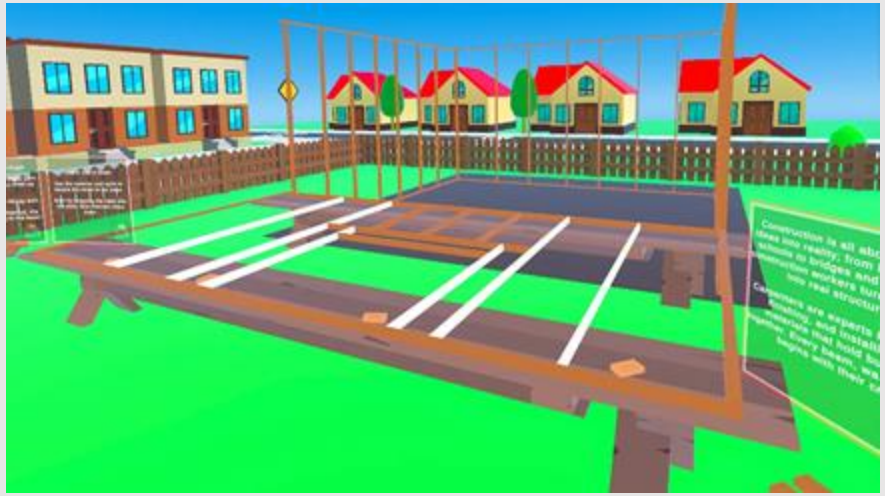
# Problem and Opportunity

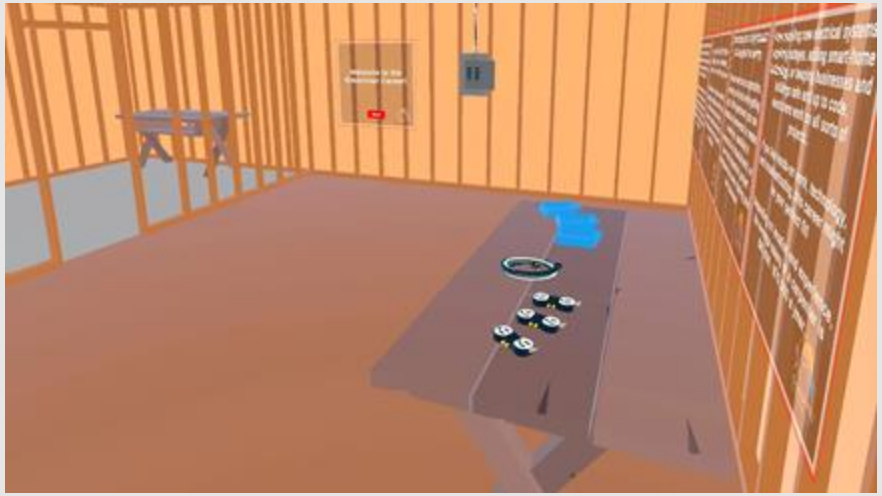
- Many High School students currently lack exposure to skilled trades and other non-traditional career paths.
- Misconceptions exist regarding accessibility and value.
- Demand for these skilled trades are ever-increasing.
- Simulations do exist in a pay-to-play model, but none offer a complete immersive experience.
- Our vision was to deliver a fully functional application for the Meta Quest 3 that allows students to explore these careers in an extended-reality environment.

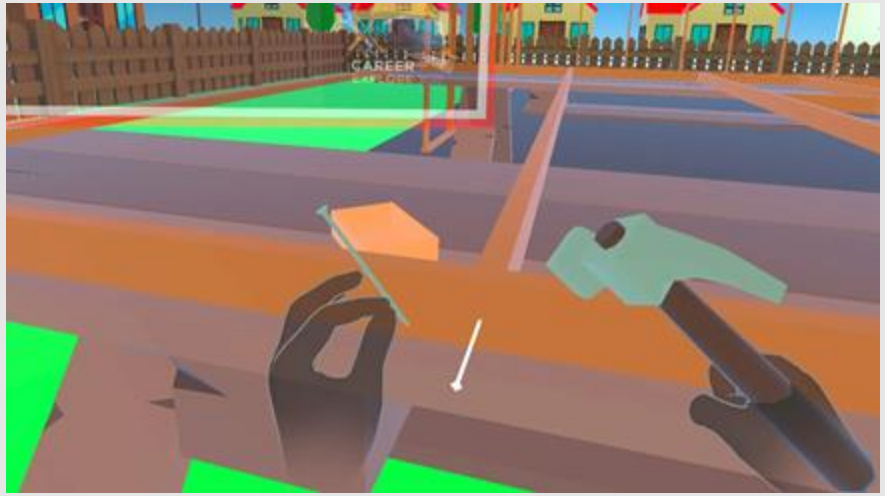
# Product Solution

- Developed with Unity using the MetaXR SDK.
- Classroom-themed Hub world.
- 4 Unique Career Scenes (Construction, Electrical, Plumbing, HVAC) with information & an activity.
- Full Hand-Tracking Compatibility.
- Available on Desktop and Mobile platforms.

# Product Solution







# Product Solution

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careerexplorexr.com

Meta Store Listing

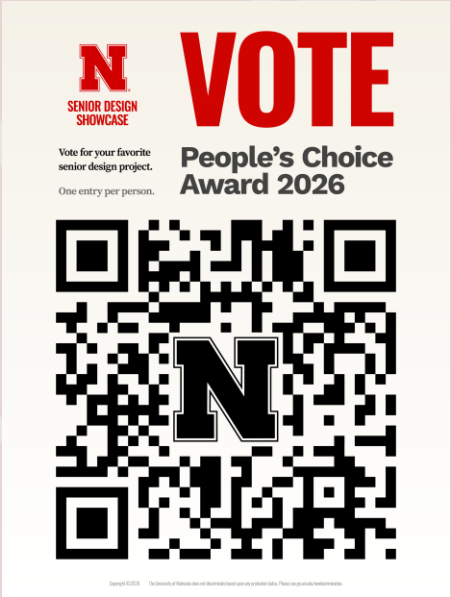
# Value and Highlights

- For the Sponsor:
  - An immersive experience designed to provide students with important and accurate information on skilled trades, as well as engaging and realistic activities.
  - An experience that is highly accessible; great performance in VR, hand-tracking, versions on other platforms.
  - Opportunities for expansion for future teams.

# Value and Highlights

- For the Student Team:
  - Valuable experience working on a large scale project utilizing industry tools, methods, and processes.
  - Important learning opportunities – managing and adapting to sponsor expectations, setting (and adhering to) realistic goals, preventing scope creep

# Thank You!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